

Assignment 3

1 Neural Machine Translation with RNNs (45 points)

- (g) *enc_mask* marks padding tokens with 1s and other non-padding tokens with 0s. And the attention scores of tokens where *enc_mask* is 1 are set to $-\infty$. Therefore, with *enc_mask*, model does not attend to padding tokens since their attention values are 0 after softmax. Such masks are necessary for NMT models to attend to meaningful tokens of a sentence, rather than padding tokens that are manually added to support parallel processing.
- (h) Corpus BLEU: 19.915406403991692.
- (i) Pros of dot production attention: computationally effective.
Cons of dot production attention: assumes the two vectors are in the same space (aligned).
Pros of additive attention: adds non-linearity and is more expressive.
Cons of additive attention: more computational overhead.

2 Analyzing NMT Systems (25 points)

- (a) The convolution layer after embedding combines individual token embeddings, which gives model better semantic representations of phrases consisting of multiple tokens.
- (b)
 - i. The NMT model translates the phrase 贼人 into the singular form *the culprit was* while the gold reference is the plural form *the culprits were*. The error occurs due to morphological ambiguity in the source language (Chinese), where the noun 贼人 does not explicitly indicate grammatical number (singular vs. plural). The model likely defaults to the singular form based on its frequency in the training data, as it lacks sufficient contextual clues or explicit number markers to infer the plural sense. To fix such errors, nouns in the source corpus could be pre-processed with explicit entity number tags.